

SUMMARY

Results-driven Mechanical Engineer with 6+ years of experience in structural analysis, product design, and cross-functional project delivery. Proven track record of reducing manufacturing costs and improving system reliability in aerospace and civil infrastructure environments. Strong communicator with a passion for sustainable engineering solutions.

EXPERIENCE

Senior Mechanical Engineer

Aerotek Engineering Solutions | Sydney, NSW | Mar 2021 – Present

- Led structural analysis of aircraft fuselage components using FEA tools (ANSYS, Abaqus), reducing material waste by 18%.
- Managed a team of 4 junior engineers on the redesign of hydraulic actuator systems for the A320 maintenance program.
- Developed and maintained technical documentation compliant with AS9100D and EASA Part 21 standards.
- Collaborated with supply chain to source ISO-certified titanium alloys, cutting lead time by 3 weeks per production cycle.

Mechanical Engineer

BridgeCo Infrastructure | Melbourne, VIC | Jan 2019 – Feb 2021

- Performed load analysis and fatigue modelling for pedestrian bridge designs across 12 government-funded projects.
- Produced detailed CAD drawings in SolidWorks and AutoCAD; coordinated with civil team on foundation interfaces.
- Conducted site inspections and wrote inspection reports for AS 3600 compliance across 8 construction sites.
- Reduced rework rate by 22% through introduction of a design review checklist adopted company-wide.

Graduate Engineer

Precision Dynamics Pty Ltd | Brisbane, QLD | Feb 2018 – Dec 2018

- Assisted in design and prototyping of industrial conveyor systems for mining clients in the Pilbara region.
- Ran thermal simulations to validate cooling system designs for heavy machinery enclosures.
- Supported senior engineers in client presentations and tender documentation.

EDUCATION

Bachelor of Engineering (Mechanical) — Honours

University of New South Wales (UNSW) | Sydney, NSW | 2014 – 2017

GPA: 6.4 / 7.0 | Thesis: Fatigue Life Prediction of Welded Steel Joints Under Variable Amplitude Loading

Certificate IV in Project Management Practice

TAFE NSW | 2020

SKILLS

CAD/CAE: SolidWorks, AutoCAD, CATIA V5, ANSYS, Abaqus **Programming:** Python, MATLAB **Standards:** AS 3600, AS9100D, ISO 9001, EASA Part 21 **Other:** GD&T, FEA, Lean Manufacturing, MS Project

PROJECTS

Solar-Powered Water Pump — Personal Project (2022)

- Designed and fabricated a low-cost solar-powered irrigation pump for a community garden in Western Sydney.
- Handled full design cycle: requirements, CAD modelling, material selection, build, and field testing.

Formula SAE — UNSW Racing Team (2016–2017)

- Contributed to suspension geometry design and upright manufacturing for the competition vehicle.
- Team placed 3rd in dynamic events at the Australasian Formula SAE competition, Melbourne 2017.

CERTIFICATIONS

- Engineers Australia — MIEAust (Chartered, 2022)
- NDIS Worker Screening Clearance (2021)
- White Card — General Construction Induction (2018)
- First Aid Certificate — St John Ambulance (2023)